

**PRE-MEDICAL : ENTHUSIAST ADVANCE COURSE PHASE - MEC-I (B)**

IMPORTANT NOTE : Students having 8 digits **Form No.** must fill two zero before their **Form No.** in OMR. For example, if your **Form No.** is 12345678, then you have to fill **0012345678**.

Test Booklet Code

This Booklet contains **28** pages.**E3**

Do not open this Test Booklet until you are asked to do so.

Important Instructions :

- The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on ORIGINAL Copy carefully with **blue/black** ball point pen only.
- The test is of **3 hours 20 minutes** duration and the Test Booklet contains **200** multiple-choice questions (four options with a single correct answer) from **Physics, Chemistry and Biology (Biology-I and Biology-II)**. **50** questions in each subject are divided into **two Sections (A and B)** as per details given below :
 - Section A** shall consist of **35 (Thirty-five)** Questions in each subject (Question Nos - 1 to 35, 51 to 85, 101 to 135 and 151 to 185). All questions are compulsory.
 - Section B** shall consist of **15 (Fifteen)** questions in each subject (Question Nos - 36 to 50, 86 to 100, 136 to 150 and 186 to 200). In Section B, a candidate needs to **attempt any 10 (Ten)** questions out of **15 (Fifteen)** in each subject. **Candidates are advised to read all 15 questions in each subject of Section B** before they start attempting the question paper. In the event of a candidate attempting more than ten questions, **the first ten questions answered by the candidate shall be evaluated.**
- Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one mark** will be deducted from the total scores. **The maximum marks are 720.**
- Use **Blue/Black Ball Point Pen only** for writing particulars on this page/markings responses on Answer Sheet.
- Rough work is to be done in the space provided for this purpose in the Test Booklet only.
- On completion of the test, the candidate **must hand over the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator** before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
- The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Form No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
- Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
- Each candidate must show on-demand his/her Allen ID Card to the Invigilator.
- No candidate, without special permission of the Invigilator, would leave his/her seat.
- The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet **twice**. **Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case.**
- Use of Electronic/Manual Calculator is prohibited.
- The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination.
- No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.**
- The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.
- Compensatory time of one hour five minutes will be provided for the examination of three hours and 20 minutes duration, whether such candidate (having a physical limitation to write) uses the facility of scribe or not.

Name of the Candidate (in Capitals) : _____

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Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ Invigilator's Signature : _____

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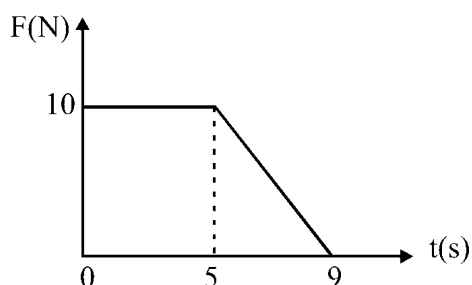
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SECTION-A (PHYSICS)

1. Ratio of weight of a man in a stationary lift and weight of a man in a lift moving downward with an acceleration is 3 : 2 then acceleration of lift is—

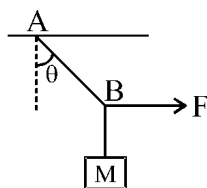
- (1) $g/3$ (2) $g/2$
(3) g (4) $2g$

2. A body of mass 4 kg is acted on by a force which varies as shown in the graph below. The momentum acquired is



- (1) 280 N-s (2) 140 N-s
(3) 70 N-s (4) 210 N-s

3. If the system is in equilibrium then tension in the string AB is



- (1) $F \sin \theta$ (2) $F/\sin \theta$
(3) $F \cos \theta$ (4) $\frac{F}{\cos \theta}$

4. A body is sliding down an inclined plane (angle of inclination 45°). If the coefficient of friction is 0.5 and $g = 9.8 \text{ m/s}^2$, then the acceleration of the body downwards in m/s^2 is :-

- (1) $\frac{4.9}{\sqrt{2}}$ (2) 4.9
(3) $\frac{19.6}{\sqrt{2}}$ (4) 19.6

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5. The depth from the surface of the earth at which the effective value of acceleration due to gravity is $\frac{g}{4}$ is (R_e = radius of earth)

- (1) R_e (2) $\frac{3R_e}{4}$ (3) $\frac{R_e}{2}$ (4) $\frac{R_e}{4}$

6. If gravitational force between planet and the satellite depends on $\frac{1}{r^n}$ where r is orbital radius. If v_0 is orbital velocity then :-

- (1) $v_0 \propto \frac{1}{r^{n-1}}$
(2) $v_0 \propto \frac{1}{\sqrt{r^{n-1}}}$
(3) $v_0 \propto \sqrt{\frac{1}{r^n}}$
(4) $v_0 \propto \frac{1}{r^n}$

7. The escape velocity of a body from the earth depends on :-

- (i) the mass of the body.
(ii) the location from where it is projected.
(iii) the direction of projection.
(iv) the height of the location from where the body is launched.

- (1) (i) and (ii) (2) (ii) and (iv)
(3) (i) and (iii) (4) (iii) and (iv)

8. Find the distance from center of a uniform solid sphere at which the gravitational potential is half of the potential at center of sphere.
(R = Radius of sphere)

- (1) $\frac{R}{2}$ (2) $\frac{R}{3}$ (3) $\frac{4R}{5}$ (4) $\frac{4R}{3}$

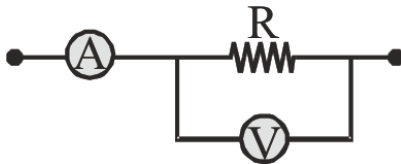
9. Near the earth's surface time period of a satellite is 1.4 hrs. Find its time period if it is at the distance '4R' from the centre of earth :-

- (1) 32 hrs. (2) $\left(\frac{1}{8\sqrt{2}}\right)$ hrs.
(3) $8\sqrt{2}$ hrs. (4) 16 hrs.

10. Two persons hold a rope of negligible weight tightly at its ends so that it is horizontal. A 15 kg weight is attached to the rope at the mid point which is now no longer remains horizontal. The minimum tension required to completely straighten the rope is

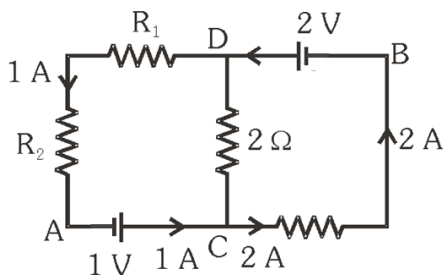
- (1) 15 kg
- (2) 15/2 kg
- (3) 5 kg
- (4) Infinitely large

11. In the circuit shown the readings of ammeter and voltmeter are 4A and 20 V respectively. The meters are non ideal, then R is :-



- (1) Equal to 5Ω
- (2) Less than 5Ω
- (3) Greater than 5Ω
- (4) Between 4Ω and 5Ω

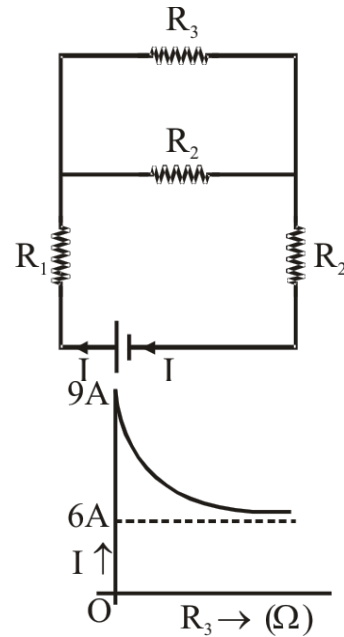
12. In the circuit shown in the figure, if the potential at point A is taken to be zero, the potential at point B is :-



- (1) +1 V
- (2) -1 V
- (3) +2V
- (4) -2 V

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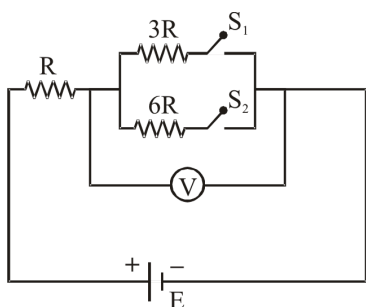
13. In the circuit shown R_3 is a variable resistance. As value of R_3 is changed, current I through the cell varies as shown in graph, then value of R_1 and R_2 are, respectively :-



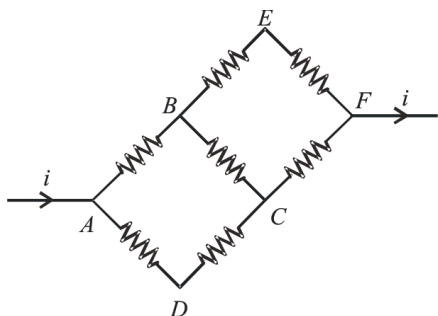
- (1) $R_1 = 2R_2$
 - (2) $2R_1 = R_2$
 - (3) $R_1 = R_2$
 - (4) None of these
14. For a cell, the terminal potential difference is 2.2 V when circuit is open, which reduces to 1.8 V when it is connected to a resistance of $R = 5 \Omega$; then internal resistance of the cell is :-
- (1) $\frac{10}{9} \Omega$
 - (2) $\frac{9}{10} \Omega$
 - (3) $\frac{11}{9} \Omega$
 - (4) $\frac{5}{9} \Omega$
15. The current density across a cylindrical conductor of radius R varies in magnitude according to the equation $J = J_0 \left(\frac{r}{R} \right)$ where r is the distance from the central axis. The current in terms of J_0 and conductor's cross-sectional area A is :-

- (1) $\frac{J_0 A}{3}$
- (2) $\frac{J_0 A}{6}$
- (3) $\frac{2J_0 A}{3}$
- (4) $\frac{J_0 A}{5}$

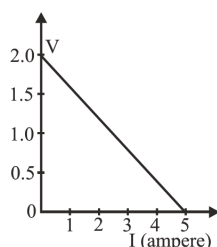
16. In the circuit shown in figure reading of voltmeter is V_1 when only S_1 is closed, reading of voltmeters V_2 when only S_2 is closed and reading of voltmeter is V_3 when both S_1 and S_2 are closed. Then :



- (1) $V_3 > V_2 > V_1$ (2) $V_2 > V_1 > V_3$
 (3) $V_3 > V_1 > V_2$ (4) $V_1 > V_2 > V_3$
17. In the adjoining circuit diagram each resistance is of $10\ \Omega$. The current in the arm AD will be



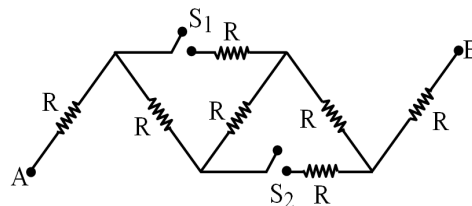
- (1) $\frac{2i}{5}$ (2) $\frac{3i}{5}$ (3) $\frac{4i}{5}$ (4) $\frac{i}{5}$
18. For a cell, the graph between the potential difference (V) across the terminals of the cell and the current (I) drawn from the cell is shown in the figure. the emf and the internal resistance of the cell are :-



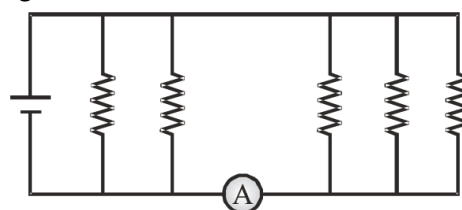
- (1) 2 V, $0.5\ \Omega$ (2) 2 V, $0.4\ \Omega$
 (3) 3 V, $0.5\ \Omega$ (4) 3 V, $0.4\ \Omega$
19. The length of a given cylindrical wire is increased by 100%. Due to the consequent decrease in diameter the change in the resistance of the wire will be :-

TC (1) 300% (2) 200% (3) 100% (4) 50%

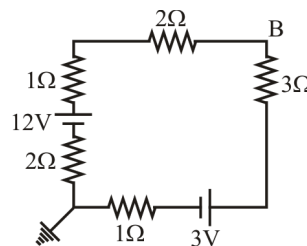
20. Find the ratio of resistance between A & B before and after switch (S_1 & S_2) are closed in the given circuit :-



- (1) 5 : 4 (2) 4 : 5 (3) 5 : 3 (4) 5 : 6
21. Five identical resistors, each of value $1100\ \Omega$ are connected to a 220 V battery as shown. The reading of the ideal ammeter A is :-



- (1) $1/5\text{ A}$ (2) $2/5\text{ A}$ (3) $3/5\text{ A}$ (4) $4/5\text{ A}$
22. Potential of the point B in the circuit below is :-

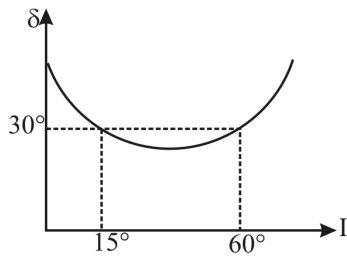


- (1) 5 V (2) 6 V (3) 7 V (4) 8 V
23. Dimensions of a block are $1\text{ cm} \times 1\text{ cm} \times 100\text{ cm}$. If specific resistance of its material is $3 \times 10^{-7}\text{ ohm-m}$, then the resistance between the opposite rectangular faces is :-

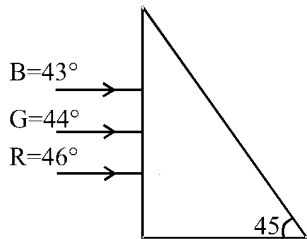
- (1) $3 \times 10^{-9}\text{ ohm}$ (2) $3 \times 10^{-7}\text{ ohm}$
 (3) $3 \times 10^{-5}\text{ ohm}$ (4) $3 \times 10^{-3}\text{ ohm}$
24. A ray of light travelling in a transparent medium of refractive index μ , falls on a surface separating the medium from air at an angle of incidence of 45° . For which of the following value of μ the ray can undergo total internal reflection ?

- (1) $\mu = 1.33$ (2) $\mu = 1.4$
 (3) $\mu = 1.5$ (4) $\mu = 1.25$

25. Figure shows the graph of angle of deviation δ versus angle of incidence i for a light ray striking a prism. The prism angle is



- (1) 30° (2) 45° (3) 60° (4) 75°
26. Figure shows a mixture of blue, green & red colored rays incident normally on a right angled prism. The critical angles of the material of the prism for red, green & blue are 46° , 44° , and 43° respectively. The arrangement will separate :-

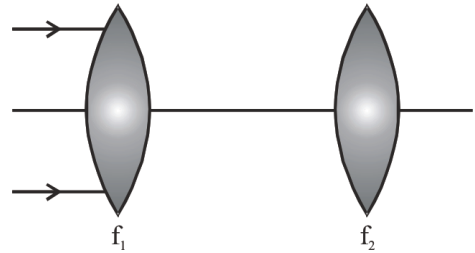


- (1) Green color from red & blue
(2) All the three colors
(3) Red color from blue & green
(4) Blue color from red & green
27. A convex lens has a focal length f . It is cut into two parts along the dotted line as shown in the figure. The focal length of each part will be :-



- (1) $\frac{f}{2}$ (2) f
(3) $\frac{3}{2}f$ (4) $2f$

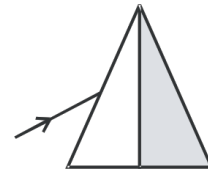
28. A parallel beam of light is incident on a system of two convex lenses of focal lengths $f_1 = 20$ cm and $f_2 = 10$ cm.



- What should be the distance between the two lenses so that rays after refraction from both the lenses pass undeviated?

- (1) 60 cm (2) 30 cm (3) 90 cm (4) 40 cm
29. A thin prism P_1 with angle 4° & refractive index 1.54 is combined with another prism P_2 of refractive index 1.72 to produce dispersion without deviation. The angle of prism P_2 is :-

- (1) 5.33° (2) 4° (3) 2.6° (4) 3°
30. A light ray is incident upon a prism in minimum deviation position and suffers a deviation of 34° . If the shaded half of the prism is knocked off, the ray will?



- (1) Suffer a deviation of 34°
(2) Suffer a deviation of 68°
(3) Suffer a deviation of 17°
(4) Not come out of the prism
31. Refractive index of a denser medium is 2, and wavelength of light in this medium is $3000\text{\AA}$ then wavelength of light in vacuum will be-

- (1) $6000\text{\AA}$
(2) $3000\text{\AA}$
(3) $1500\text{\AA}$
(4) $4500\text{\AA}$

32. Two rays made of red colour & blue colour are incidenting on a plane mirror with same angle of incidence then-

- (1) Blue colour will deviate more than red
- (2) Red colour will deviate more than blue
- (3) Both colour will deviate with same angle
- (4) It depends on medium

33. A ray of light is incidenting at angle 45° in air and passing in the denser medium of refractive index $\sqrt{2}$ then angle of deviation-

- (1) 45° (2) 30° (3) 60° (4) 15°

34. An object of height +5 mm is placed $\perp$ to principal axis at 60 cm away from concave lens of focal length 60 cm then height of its image will be-

- (1) -2.5 mm (2) +2.5 mm
- (3) 10 mm (4) infinite

35. Power of convex and concave lenses is 5D. These lenses are place in contact with each other then focal length of combination will be-

- (1) Infinite (2) Zero
- (3) 10 cm (4) 20 cm

SECTION-B (PHYSICS)

36. A particle of mass M is situated at centre of a spherical shell of same mass and radius a. The gravitational potential at a point situated at $a/2$ distance from the centre will be

- (1) $\frac{-3GM}{a}$ (2) $\frac{-2GM}{a}$
- (3) $\frac{-GM}{a}$ (4) $\frac{-4GM}{a}$

37. If radius of earth decreases by 1% but its mass remains constant, then the value of acceleration due to gravity on earth surface increases by :

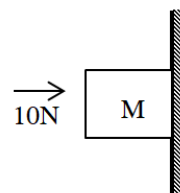
- (1) 2 % (2) 1 % (3) 3 % (4) 1/2 %

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38. A vehicle of 100 kg is moving with a velocity of 5 m/sec. To stop it in $1/10$ sec, the required force in opposite direction is :

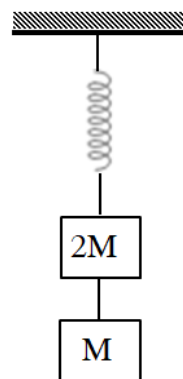
- (1) 5000 N (2) 50 N
- (3) 500 N (4) 1000 N

39. A horizontal force of 10 N is necessary to just hold a block stationary against a wall. The coefficient of friction between the block and wall is 0.2, then weight of the block is :



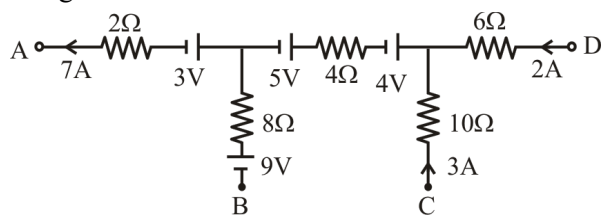
- (1) 2 N (2) 4 N (3) 20 N (4) 40 N

40. The given system is at rest, find acceleration of both the blocks just after the rope is cut :



- (1) $\frac{g}{2}, g$ (2) $2g, g$
- (3) g, g (4) $\frac{g}{2}, \frac{g}{2}$

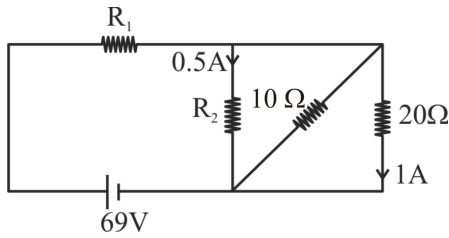
41. In a portion of some large electrical network, currents in certain branches are known as shown in figure :-



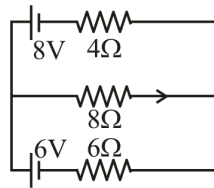
The value of $V_A - V_C$ is

- (1) 76 V (2) -76 V
- (3) -58 V (4) -52 V

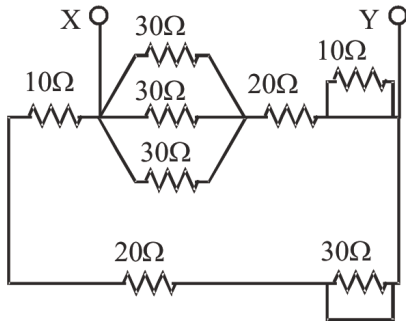
42. In the circuit shown in the given figure the resistance R_1 and R_2 are respectively



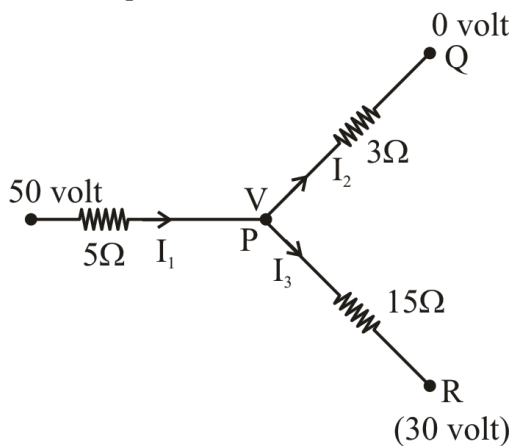
- (1) $14\ \Omega$ and $40\ \Omega$ (2) $40\ \Omega$ and $14\ \Omega$
 (3) $40\ \Omega$ and $30\ \Omega$ (4) $14\ \Omega$ and $30\ \Omega$
43. The current in $8\ \Omega$ resistance is : (as per given circuit)



- (1) 0.69 A (2) 0.92 A
 (3) 1.30 A (4) 1.6 A
44. Net resistance between X and Y is –



- (1) $5\ \Omega$ (2) $10\ \Omega$ (3) $15\ \Omega$ (4) $60\ \Omega$
45. For the given circuit as shown in figure. The value of V at point P :-

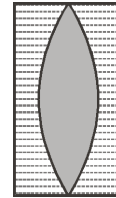


- (1) 10 V (2) 20 V (3) 5 V (4) 7.5 V

46. A ray of light travelling horizontally is incident on a transparent sphere ($\mu = \sqrt{3}$). If it emerges from the other end of the parallel diameter the angle of incidence will be :-

- (1) 30° (2) 60°
 (3) $\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$ (4) $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$

47. As shown in figure, a convergent lens is placed inside a cell filled with liquid. The lens has focal length + 20 cm when in air, and its material has refractive index 1.50. If the liquid has refractive index 1.60, the focal length of the system is :



- (1) +80 cm (2) -80 cm
 (3) -24 cm (4) -100 cm
48. A ray is travelling from rare medium to denser medium then ray-
- (1) Always deviates towards normal
 (2) Always deviates away from normal
 (3) May have deviation angle zero
 (4) Always has angle of deviation zero
49. If refractive index of denser medium is very large. A ray is travelling from air to denser medium then refracted ray will remains-
- (1) Near to normal
 (2) Near to boundary separating two mediums
 (3) Undeviated
 (4) Closer to incidenting ray
50. A ray is incidenting on a prism of refractive index $\sqrt{2}$ and prism angle 30° with angle of incidence 45° then angle of emergence of emerging ray will be-

- (1) 30° (2) 45° (3) 15° (4) None

Topic : GOC-I, ELECTROCHEMISTRY, SOLID STATE (TILL TAUGHT), [BACK UNIT] : CHEMICAL KINETICS

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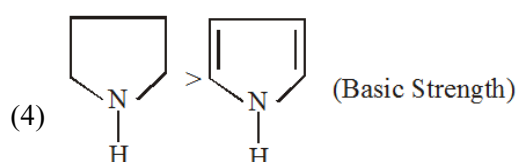
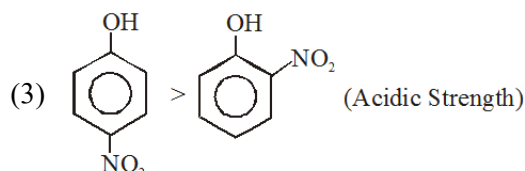
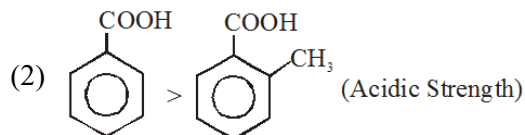
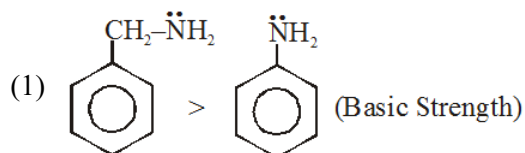
SECTION-A (CHEMISTRY)

51. With the help of given reaction :
 $2\text{MnO}_4^- + 5\text{NO}_2^- + 6\text{H}^+ \rightarrow 2\text{Mn}^{+2} + 5\text{NO}_3^- + 3\text{H}_2\text{O}$
 Calculate the rate (in Ms^{-1}) at which the NO_2^- concentration is decreasing if the MnO_4^- is decreasing at the rate of 0.024 M s^{-1} .
 (1) 4.8×10^{-3} (2) 6×10^{-2}
 (3) 3×10^{-2} (4) 2.4×10^{-3}
52. For a particular reaction, the rate expression is given as $r = k[\text{A}][\text{B}]^{0.5}$. If the volume of vessel is reduced to one-fourth of the initial volume, the rate of reaction will.
 (1) becomes 1/4 times
 (2) becomes 8 times
 (3) becomes 1/8 times
 (4) remains unaffected
53. For a first order reaction $\text{X} \rightarrow \text{Y} + \text{Z}$, at the beginning of reaction 10 gm of A were taken, then what is the amount of A left after 6 half lives ?
 (1) 1.56 gm
 (2) 0.625 gm
 (3) 0.3125 gm
 (4) 0.156 gm
54. The rate equation for the reaction; $2\text{A} + \text{B} \rightarrow \text{C}$ is rate = $K[\text{A}][\text{B}]$. The correct statement about this is :
 (1) K is independent of [A] and [B]
 (2) $t_{1/2}$ is constant
 (3) Unit of K is sec^{-1}
 (4) Rate of formation of C is twice the rate of disappearance of A
55. **Assertion (A)** : In equation $r = P Z_{\text{AB}} e^{\frac{-E_a}{RT}}$, P (steric factor) takes into account the fact that at the time of collision, reactant molecules must have threshold energy for effective collision.
Reason (R) : As per collision theory, activation energy and proper orientation of the molecules together determine the criteria for an effective collision.
 In light of above statement, select the correct option :-
 (1) A is true and R is false
 (2) A and R are correct and R is correct explanation of A.
 (3) A and R are correct but R is not correct explanation of A.
 (4) A is false but R is true.
56. The half life of a first order reaction is $6.93 \times 10^{-3} \text{ min}$ at 27°C . At this temperature, $10^{-8} \%$ of the reactant molecules are able to cross the energy barrier. The pre-exponential factor A in the Arrhenius equation is equal to :-
 (1) 10^4 min^{-1} (2) 10^8 min^{-1}
 (3) 10^{10} min^{-1} (4) 10^{12} min^{-1}
57. A 250.0 mL sample of a 0.20 M Cr^{3+} is electrolyzed with a current of 96.5 A. If remaining $[\text{Cr}^{3+}]$ is 0.1 M the duration of process is :-
 (1) 25 sec (2) 225 sec
 (3) 150 sec (4) 75 sec
58. $\text{X}^\ominus + \text{Y} \rightarrow \text{X} + \text{Y}^\ominus$
 $\text{Y}^\ominus + \text{Z} \rightarrow \text{No reaction}$
 $\text{Z}^\ominus + \text{X} \rightarrow \text{Z} + \text{X}^\ominus$
 The correct order of standard electrode potential is :-
 (1) $E_{\text{Z}^\ominus/\text{Z}}^\circ < E_{\text{X}^\ominus/\text{X}}^\circ < E_{\text{Y}^\ominus/\text{Y}}^\circ$
 (2) $E_{\text{Z}^\ominus/\text{Z}}^\circ < E_{\text{Y}^\ominus/\text{Y}}^\circ < E_{\text{X}^\ominus/\text{X}}^\circ$
 (3) $E_{\text{Z}^\ominus/\text{Z}}^\circ > E_{\text{X}^\ominus/\text{X}}^\circ > E_{\text{Y}^\ominus/\text{Y}}^\circ$
 (4) $E_{\text{Z}^\ominus/\text{Z}}^\circ > E_{\text{Y}^\ominus/\text{Y}}^\circ > E_{\text{X}^\ominus/\text{X}}^\circ$

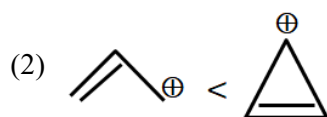
TG: @Chalnaayaaar

59. E° for $\text{Cl}_2(\text{g}) + 2\text{e}^- \rightarrow 2\text{Cl}^-(\text{aq})$ is 1.36 V; E° for $\text{Cl}^-(\text{g}) \rightarrow 1/2\text{Cl}_2(\text{g}) + \text{e}^-$ is :
- (1) 1.36 V (2) -1.36 V
(3) -0.68 V (4) 0.68 V
60. **Assertion (A)** : Cell potential of mercury cell remain constant during its life.
Reason (R) : As overall cell reaction of every primary battery does not involve an ion in solution, therefore cell potential remain constant. In the light of above statements, select the correct option –
- (1) Both **A** and **R** are correct and **R** is correct explanation of **A**.
(2) Both **A** and **R** are correct but **R** is not correct explanation of **A**.
(3) **A** is true but **R** is false
(4) **A** is false but **R** is true
61. The molar conductance at infinite dilution for AgNO_3 , AgCl and NaCl are $116.5 \text{ S cm}^2 \text{ mol}^{-1}$, $121.6 \text{ S cm}^2 \text{ mol}^{-1}$ and $110.3 \text{ S cm}^2 \text{ mol}^{-1}$ respectively. The limiting molar conductance of NaNO_3 is -
- (1) $111.4 \text{ S cm}^2 \text{ mol}^{-1}$ (2) $105.2 \text{ S cm}^2 \text{ mol}^{-1}$
(3) $130.6 \text{ S cm}^2 \text{ mol}^{-1}$ (4) $150.2 \text{ S cm}^2 \text{ mol}^{-1}$
62. The E° at 25°C for the following reaction is 0.22 V, then equilibrium constant at 25°C is -
 $\text{H}_2(\text{g}) + 2\text{AgCl}(\text{s}) \rightleftharpoons 2\text{Ag}(\text{s}) + 2\text{HCl}(\text{aq})$
- (1) 2.8×10^7 (2) 5.2×10^8
(3) 5.2×10^6 (4) 5.2×10^3
63. What is the ratio of moles of Ag, Cu and Al deposited if same amount of electricity is passed through $\text{AgNO}_3(\text{aq})$, $\text{CuSO}_4(\text{aq})$ and molten Al_2O_3 solutions :-
- (1) 1 : 1 : 1 (2) $1 : \frac{1}{2} : \frac{1}{3}$
(3) $\frac{1}{3} : \frac{1}{2} : 1$ (4) 1 : 2 : 3
64. In simple cubic unit cell, body centred cubic unit cell, number of lattice points are respectively –
- (1) 1, 2 (2) 4, 2
(3) 8, 9 (4) 9, 8
65. Which solid is also called giant molecule –
- (1) MgO (2) Ag
(3) CCl_4 (4) SiO_2
66. End centred unit cell exist in –
- (1) Cubic system
(2) Hexagonal system
(3) Monoclinic system
(4) Triclinic system
67. In a cubic solid A atoms are at alternate corners, B atoms are at alternate face centres and C atoms are at alternate edge centres and at body centres, then formula of compound is –
- (1) ABC_2 (2) ABC
(3) AB_2C_4 (4) A_2BC_4
68. Mass of Cu which deposits at cathode during electrolysis of $\text{CuSO}_4(\text{aq.})$ solution, if 1.12 L $\text{O}_2(\text{g})$ is evolved at anode at 1 atm and 273 K under similar conditions - (Atomic weight of Cu = 63.5 amu)
- (1) 1.12 g
(2) 2.24 g
(3) 6.35 g
(4) 63.5 g
69. Following data is obtained for reaction
 $\text{A}(\text{g}) \rightarrow \text{B}(\text{g}) + \text{C}(\text{g})$
- | | | | | |
|----------------|---|-----|------|-------|
| Time (in min.) | 0 | 10 | 20 | 30 |
| P_A (in atm) | 2 | 1.8 | 1.62 | 1.458 |
- then order of reaction is –
- (1) 0 (2) 1
(3) 2 (4) 3

78. Which is the incorrect order :-



79. Which of the following is correct order of resonance energy :



(4) All are correct

80. **Statement-I** : Acetic acid is stronger acid than phenol

Statement-II : Conjugate base of acetic acid is less stable than conjugate base of phenol

(1) Statement-I is correct and Statement-II is incorrect

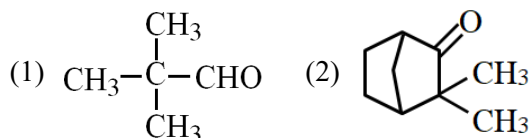
(2) Statement-I and Statement-II both are correct

(3) Statement-I and Statement-II both are incorrect

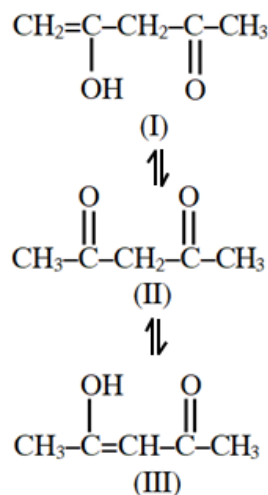
(4) Statement-I is incorrect and Statement-II is

TG: @Chalnaayaaar

81. Which of the following show tautomerism ?



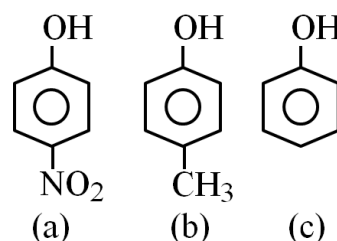
82. Correct stability order of given forms is



(1) II > III > I (2) I > II > III

(3) III > II > I (4) II > I > III

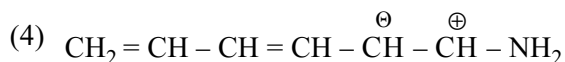
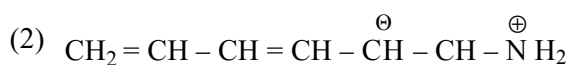
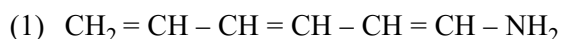
83. Correct acidic strength order is



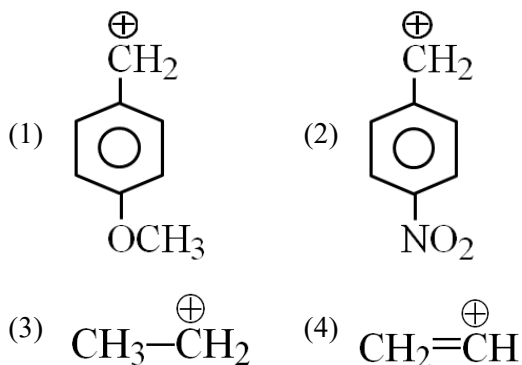
(1) c > a > b (2) a > b > c

(3) a > c > b (4) c > b > a

84. Which of the following resonating structure is least stable :-

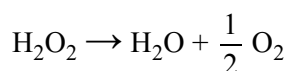


85. Most stable cation is



SECTION-B (CHEMISTRY)

86. Which rate equation is correct for reaction



- (1) $-\frac{d[\text{H}_2\text{O}_2]}{dt} = \frac{d[\text{H}_2\text{O}]}{dt} = \frac{1}{2} \frac{d[\text{O}_2]}{dt}$
 (2) $-\frac{d[\text{H}_2\text{O}_2]}{dt} = \frac{1}{2} \frac{d[\text{H}_2\text{O}]}{dt} = \frac{d[\text{O}_2]}{dt}$
 (3) $-\frac{1}{2} \frac{d[\text{H}_2\text{O}_2]}{dt} = \frac{d[\text{H}_2\text{O}]}{dt} = 2 \frac{d[\text{O}_2]}{dt}$
 (4) $-\frac{d[\text{H}_2\text{O}_2]}{dt} = \frac{d[\text{H}_2\text{O}]}{dt} = 2 \frac{d[\text{O}_2]}{dt}$

87. The ratio of time required for $7/8^{\text{th}}$ of the reaction of first order to complete to the time required for half of the reaction to complete :-

- (1) 3 : 1 (2) 1 : 3 (3) 1 : 2 (4) 2 : 1

88. The resistance of 0.01 N NaCl solution at 25°C is 200Ω . Cell constant of conductivity cell is 1 cm^{-1} . The equivalent conductance is :-

- (1) $5 \times 10^2 \Omega^{-1} \text{ cm}^2 \text{ eq}^{-1}$
 (2) $6 \times 10^3 \Omega^{-1} \text{ cm}^2 \text{ eq}^{-1}$
 (3) $7 \times 10^4 \Omega^{-1} \text{ cm}^2 \text{ eq}^{-1}$
 (4) $8 \times 10^5 \Omega^{-1} \text{ cm}^2 \text{ eq}^{-1}$

89. The conductivity of strong electrolyte.

- (1) increase on dilution
 (2) decrease on dilution
 (3) does not change with dilution
 (4) maximum at infinite dilution

TC: @Chalnaqyaaar

90. The limiting molar conductance of Al^{3+} & SO_4^{2-} are 81 and $90 \text{ Ohm}^{-1} \text{ cm}^2 \text{ mol}^{-1}$ respectively. Then what will be $\Lambda_{\text{eq}}^\infty$ of $\text{Al}_2(\text{SO}_4)_3$?

- (1) $72 \text{ S cm}^2 \text{ geq}^{-1}$ (2) $171 \text{ S cm}^2 \text{ geq}^{-1}$
 (3) $110 \text{ S cm}^2 \text{ geq}^{-1}$ (4) $432 \text{ S cm}^2 \text{ geq}^{-1}$

91. What is the potential of the cell containing two hydrogen electrodes as represented below
 $\text{Pt}; \text{H}_2(\text{g}) | \text{H}^+(10^{-8} \text{ M}) || \text{H}^+(0.001 \text{ M}) | \text{H}_2(\text{g}) | \text{Pt}$ (s);

- (1) -0.295 V (2) -0.0591 V
 (3) 0.295 V (4) 0.0591 V

92. Xe crystallises in FCC lattice, if radius of Xe atom is around 220 pm , then edge length of unit cell and coordination number of Xe is respectively -

- (1) 622 pm , 8 (2) 440 pm , 8
 (3) 622 pm , 12 (4) 440 pm , 12

93. For reaction $\text{X} + 2\text{Y} \rightarrow \text{Z}$, rate law is $-\frac{d[\text{X}]}{dt} = K[\text{X}][\text{Y}]^2$, if following data is given

	$[\text{X}]_0 \times 10^{-3} \text{ M}$	$[\text{Y}]_0 \times 10^{-3} \text{ M}$	Half life
1	3	200	40 sec.
2	6	200	t sec.

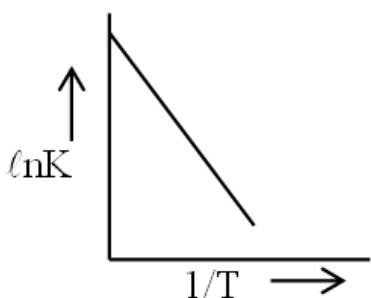
then t is-

- (1) 80 (2) 40 (3) 160 (4) 360

94. Select the correct statement -

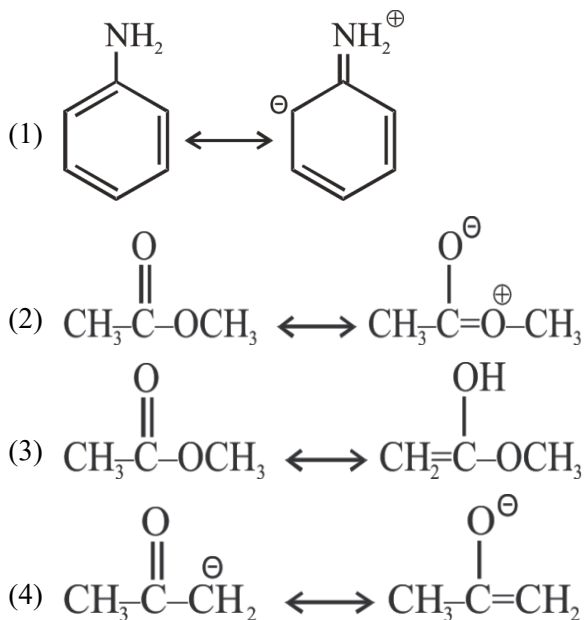
- (1) Half life of a second order reaction is dependent of initial concentration of reactant.
 (2) A first order reaction must be a complex reaction
 (3) Reaction $2\text{A}(\text{g}) + 5\text{B}(\text{g}) \rightarrow 4\text{C}(\text{g})$ can be single step reaction.
 (4) During the course of reaction, rate of reaction always decreases.

95. In the given plot between $\ln K$ and $\frac{1}{T}$

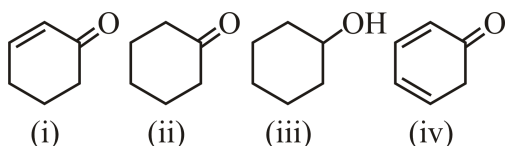


intercept and slope are respectively–

- (1) $\log A, \frac{-E_a}{2.303R}$ (2) $\ln A, \frac{-E_a}{2.303R}$
 (3) $\ln A, \frac{-E_a}{R}$ (4) $\log A, \frac{-E_a}{R}$
96. Which of the following is INVALID canonical form :



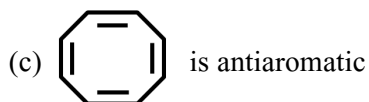
97. Correct order of (C–O) bond length for the following set of molecules –



- (1) (iii) > (iv) > (i) > (ii)
 (2) (ii) > (i) > (iv) > (iii)
 (3) (iii) > (i) > (iv) > (ii)
 (4) (ii) > (iv) > (i) > (iii)

TG: @Chalnaayaaar

98. Select correct statement(s)
- (a) Electromeric effect is temporary effect
 (b) If Inductive effect and Electromeric effect operates in opposite direction then electromeric effect dominates.

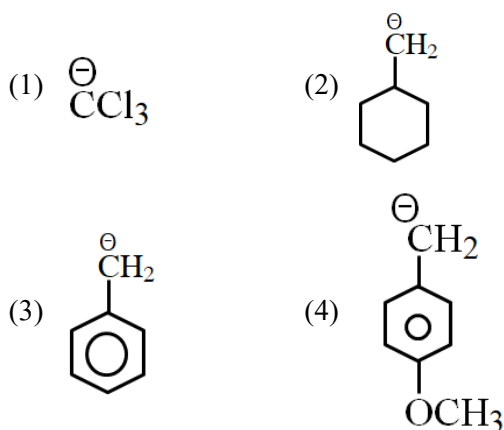


- (1) a,b,c
 (2) a,b
 (3) a,c
 (4) b,c

99. Final rearranged cation of

- (1)
 (2)
 (3)
 (4) No rearrangement

100. Most stable anion is



Topic : ORIGIN & EVOLUTION (FROM THEORY OF ORGANIC EVOLUTION), HUMAN REPRODUCTION (UPTO MENSTRUAL CYCLE), ECOLOGY (BIODIVERSITY, POLLUTION), LOA (UPTO FEMALE REPRODUCTIVE SYSTEM), [BACK UNIT] : HUMAN HEALTH AND DISEASE

SECTION-A (BIOLOGY-I)

101. The organization which publishes the 'Red Data Book' is:-

- (1) National Environment Engineering Research Institute
- (2) International Union of Conservation of Nature and Natural Resources
- (3) ICBN
- (4) National Wild life Action Plan

102. Match the column I and coloumn II

	Column-I		Column-II
(a)	Dodo	(i)	Africa
(b)	Quagga	(ii)	Bali, Javan, Caspian
(c)	Thylacine	(iii)	Mauritius
(d)	Steller's Sea Cow	(iv)	Australia
(e)	Tiger sub species	(v)	Russia

- (1) (a)- iv (b)- ii (c)- i (d)- v (e) -iii
- (2) (a)- iii (b)- iv (c)- ii (d)- i (e) -iii
- (3) (a)- iii (b)- i (c)- iv (d)- v (e) -ii
- (4) (a)- iii (b)- v (c)- iv (d)- i (e) -ii

103. Threats to biodiversity comes from :-

- (1) Habitat loss
- (2) Over exploitation
- (3) Intensive (excessive) agriculture
- (4) All of these

104. Which one of the following shows maximum genetic diversity in India ?

- (1) Mango
- (2) Groundnut
- (3) Rice
- (4) Maize

TC: @Chahuaa

105. Which of the following is a property of Hot Spot ?

- (1) High Temperature
- (2) High degree of endemism
- (3) Direct economic benefits are derived
- (4) Presence of deserts

106. Which of the following are the last refuges for rare and endangered plants ?

- (1) Keoladev national park
- (2) Sacred groves
- (3) Zoological parks
- (4) Botanical gardens

107. The World Summit at Johannesburg was held with the theme of ?

- (1) Conservation of energy
- (2) Sustainable developement
- (3) Saving natural resources
- (4) Saving animals in wild

108. Given below are two statements which are labelled as :

Assertion (A) : Electrostatic precipitator, which can remove over 99 percent particulate matter present in the exhaust.

Reason (R) : The particulate size 2.5 micrometers or less in diameter are responsible for causing the greatest harm to human health.

- (1) Both (A) and (R) are correct and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- (3) (A) is not correct but (R) is correct
- (4) Both (A) and (R) incorrect

109. Assertion: Discharging sewage in river leads to increase in BOD of water.

Reason: Sewage contains suspended impurities.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

110. Which of the following is not associated with increase in atmospheric concentration of CO_2 ?

- (1) Increase in mean annual temperature
- (2) Increase in sea level
- (3) Expected increase in productivity of C_3 plants
- (4) Increase in the rate of ozone depletion

111. Euro-IV are emission norms for reducing :-

- (1) O_3 and CO_2
- (2) NO_2 and N_2O
- (3) Sulphur and aromatic compounds
- (4) CO_2 and particulate matter

112. Which of the following is not one of the prime health risks with greater UV radiation through the atmosphere due to depletion of stratospheric ozone?

- (1) Reduced immune system
- (2) Damage to eyes
- (3) Increased liver cancers
- (4) Increased skin cancer

113. "Good ozone" is found in the :-

- | | |
|------------------|-----------------|
| (1) Stratosphere | (2) Ionosphere |
| (3) Mesosphere | (4) Troposphere |

TG: @Chalnaayaaar

114. Ozone is gaseous "secondary" air pollutant because :-

- (1) It is not an important pollutant
- (2) It causes secondary effects on plants and animals
- (3) It is formed by the oxides of nitrogen and hydrocarbons in sunlight and warm temperature
- (4) It is formed by the oxidation of oxygen in stratosphere

115. Assertion :- From one functional megaspore 7 celled and 8 nucleated embryo sac is developed.

Reason :- Formation of megaspores from the megaspore mother cell is called megasporonogenesis.

- (1) Both Assertion & Reason are True and the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

116. Assertion : Filliform apparatus guides the pollen tube into the synergids.

Reason : It is special cellular thickening at micropylar tip that secrete chemotropic stimulus.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

117. Which of the following statements about sporopollenin is false ?

- (1) No enzyme that degrades sporopollenin so far is known.
- (2) Sporopollenin is one of the resistant organic material.
- (3) Sporopollenin can withstand high temperature and strong acids.
- (4) Germ pores are the areas with high sporopollenin content.

118. In the fully organised normal Embryosac, what is the number of diploid cells?
(1) 2 (2) 3 (3) 0 (4) 4
119. Algal bloom forming algae are extremely toxic to -
(1) Fishes (2) Humans
(3) Animals (4) All of these
120. How many plants in the list given below have many ovules in an ovary?
Wheat, Paddy, Mango, Papaya, Water melon, Orchids
(1) Two (2) Three (3) One (4) Four
121. Select Incorrect statement from following ?
(1) Loss of biodiversity leads to increased variability in plant productivity.
(2) Rate of extinction in sixth episode of mass extinction is 100 to 1000 times greater than in previous five episodes.
(3) Extinctions across taxa are random.
(4) Many extinctions in last 500 years were due to overexploitation by humans.
122. What is the name of the catfish introduced in India for aquaculture purposes ?
(1) *Hilsa* (2) *Catla* (3) *Clarias* (4) *Rohu*
123. Animals that have been extinct in wild , are maintained in which of the following ?
(1) Zoo
(2) Biosphere reserve
(3) Sacred groves
(4) Wildlife sanctuaries
124. Watching flowers in full bloom and listening to Bulbul's song in spring comes under which argument for conservation of Biodiversity ?
(1) Narrow utilitarian (2) Philosophical
(3) Broad utilitarian (4) Spiritual
125. Which of the following is adopted as substitute for open burning dumps?
(1) Incineration
(2) Discarding waste in oceans
(3) Sanitary Land fills
(4) Composting
126. Which of the following is not related to hallucination ?
(1) Bhang (2) Morphine
(3) Charas (4) L.S.D
127. First mother milk provide which kind of immunity ?
(1) Active immunity
(2) Passive immunity
(3) Natural passive acquired immunity
(4) Artificial passive immunity
128. Which of the following is an allergic disease?
(1) Tetanus (2) Asthma
(3) 'Whooping laugh (4) Influenza
129. Which of the following is not a symptom of withdrawal syndrome ?
(1) Anxiety (2) Euphoria
(3) Sweating (4) Shakiness
130. Which is best technique for cancer detection ?
(1) Biopsy [histo-pathology]
(2) M.R.I.
(3) X-ray
(4) Computed tomography
131. Fluid filled alveoli is feature of :-
(1) Amoebiasis (2) Pneumonia
(3) Ascariasis (4) Typhoid

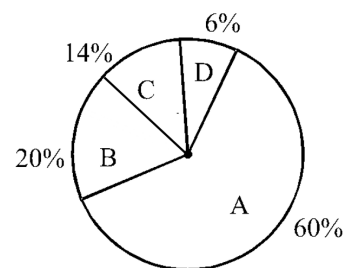
132. Rhino virus infect
- (1) Nose only
 - (2) Nose Respiratory passage
 - (3) Nose, Respiratory passage, Lungs
 - (4) Nose, Respiratory passage, Lungs, Blood
133. Malignant malaria is caused by
- (1) *P. Vivax*
 - (2) *P. Falciparum*
 - (3) *P. Ovali*
 - (4) *P. Malariae*
134. Deficiency of which enzyme, result in severe combined immunodeficiency syndrome?
- (1) Adenosine hydroxylase
 - (2) Adenosine aminase
 - (3) Adenosine deaminase
 - (4) Homogentisic acid oxidase
135. Choose the incorrect match
- (1) Down syndrome – Trisomy of 18th Chromosome
 - (2) Patau syndrome – Trisomy of 13th Chromosome
 - (3) Edward syndrome – Trisomy of 8th chromosome
 - (4) Both (1) & (3)

SECTION-B (BIOLOGY-I)

136. In your opinion, which is the most effective way to conserve the plant diversity of an area?
- (1) By creating biosphere reserve
 - (2) By creating botanical garden
 - (3) By developing seed bank
 - (4) By tissue culture method

TG: @Chalnaayaaar

137. Resistant and resilient to occasional disturbance (natural or man-made) and resistant to invasions by alien species. It is a character of :-
- (1) Unstable community with more species and high biomass & productivity
 - (2) Unstable community with less biomass or less productivity
 - (3) Stable community with more species and less biomass or less productivity
 - (4) Stable community with more species and high biomass or high productivity
138. Which of the following has accounted for greater biodiversity in tropical regions ?
- (1) Natural calamities frequently promoted succession in tropical regions
 - (2) Constant environment promoted niche overlapping that led to greater biodiversity
 - (3) More solar availability
 - (4) More fertile soil that promoted growth of producers and decomposers
139. Given pie-diagram represents the relative contribution of various green house gases to total global warming. Select the correct statement(s) regarding A, B, C & D.



- (1) A is the gas which is produced during the combustion of fossil fuel.
- (2) B are the chemicals which are used as coolants in refrigerators.
- (3) C is the gas which is major constituent of biogas.
- (4) D is the chemical which is a hydrocarbon.

140. EURO-III norms stipulate that sulphur be controlled at :-
- (1) 150 ppm in petrol and 350 ppm in diesel
 - (2) 350 ppm in petrol and 150 ppm in diesel
 - (3) 50 ppm in petrol and 50 ppm in diesel
 - (4) 50 ppm in petrol and 100 ppm in diesel
141. Global agreement in specific control strategies to reduce the release of ozone depleting substances, was adopted by :-
- (1) Kyoto Protocol
 - (2) Vienna Convention
 - (3) Rio de Janerio Conference
 - (4) Montreal Protocol
142. How many meiosis and mitotic division are required for the formation of mature male gametophyte from microspore mother cell in angiosperm ?
- (1) 1 and 1
 - (2) 1 and 2
 - (3) 2 and 1
 - (4) 2 and 2
143. What is correct about the pollen grain of flowering plants ?
- (1) Large vegetative cell with large nucleus
 - (2) Small vegetative cell with small nucleus
 - (3) Small vegetative cell with large nucleus
 - (4) Small generative cell with large nucleus
144. A flower of Mustard has six stamens, each microsporangium has 10 pollen mother cells. How many pollen grains are found in the flower :-
- (1) 450
 - (2) 2500
 - (3) 960
 - (4) 680
145. Sequence of development during the formation of embryosac is –
- (1) Archegonium → megaspore → megaspore mother cell
 - (2) Megaspore mother cell → archegonium → megaspore → embryosac
 - (3) Megaspore → megaspore mother cell → archegonium → embryosac
 - (4) Archegonium → megaspore mother cell → megaspore → embryosac
146. Find out the incorrect matching regards innate immunity :
- (1) Physiological barrier-epithelium lining of respiratory, G.I.T, urogenital tract
 - (2) Cytokines- Interferons
 - (3) Cellular - natural killer cells
 - (4) Physical - skin, mucosa
147. Which lymphoid organ act a filter of blood by trapping blood borne-microorganism ?
- (1) Bone marrow
 - (2) Lymphnode
 - (3) Spleen
 - (4) Tonsils
148. Macrophage and neutrophil participate in :-
- (1) Lymphotoxin production
 - (2) Lymphokine production
 - (3) Phagocytosis
 - (4) Acquired immunity
149. Vaccine when used as prophylaxis elicit which type of immune response?
- (1) Primary
 - (2) Secondary
 - (3) Tertiary
 - (4) Quarternary
150. Sustained high fever (39° to 40°C), weakness, stomach pain, constipation, headache, loss of appetite; all are the symptom of which disease.
- (1) Pneumonia
 - (2) Thyphoid
 - (3) Malaria
 - (4) Common cold

Topic : ORIGIN & EVOLUTION (FROM THEORY OF ORGANIC EVOLUTION), HUMAN REPRODUCTION (UPTO MENSTRUAL CYCLE), ECOLOGY (BIODIVERSITY, POLLUTION), LOA (UPTO FEMALE REPRODUCTIVE SYSTEM), [BACK UNIT] : HUMAN HEALTH AND DISEASE

SECTION-A (BIOLOGY-II)

151. Which of the following statements is true ?

- (A) Darwin explained the natural selection as directional process
 (B) Pre-adaptive advantageous mutations brings about change in phenotypes and are responsible for evolution of new species
 (C) Genetic equilibrium is possible even if sources of variations remain active

- (1) A,B,C (2) A,B
 (3) A,C (4) C only

152. Read the following four (A-D) statements :

- (A) When migration of a section of population to another place and population occurs, gene frequencies changes
 (B) If gene migration happens multiple times there would be a gene flow
 (C) If loss of gene from a population occurs by-chance it is called as genetic drift
 (D) Natural selection is a process in which heritable variations enabling better survival are enabled to reproduce greater number of progeny.
 How many of the above statements are true ?

- (1) Two (2) One
 (3) Four (4) Three

153. Genetic drift has more effect on :-

- (1) Small population (2) Large population
 (3) Biosphere (4) Species

154. The ultimate source of variations is

- (1) Sexual reproduction
 (2) Hormonal action
 (3) Natural selection
 (4) Mutations

155. Neo-Darwinism believes that new species develop through :

- (1) mutations
 (2) hybridization
 (3) mutations with natural selection
 (4) none of the above

156. Sertoli cells are help in :-

- (1) Excretion
 (2) Respiration
 (3) Nutrition of sperms
 (4) Egg formation

157. In reproductive age of female, eggs are release in the form of :-

- (1) Primary oocyte
 (2) Oogonia
 (3) Secondary Oocyte
 (4) Atretic-Follicle

158. No. of seminiferous tubule in each testicular lobule :-

- (1) 750-1000 (2) 1500-2000
 (3) 1-3 (4) 250-300

159. Loose fold of skin which cover glans-penis is known as :-

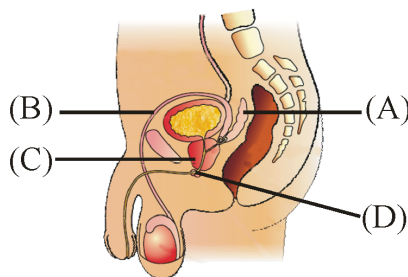
- (1) Fore-Skin
 (2) Corpus-Spongiosum
 (3) External-Genitalia
 (4) Corpora-Cavernosa

160. How many Y-chromosomes are present in Ist polar body in human being ?

- (1) 23 (2) 01 (3) 02 (4) 00

161. In human females, meiosis-I start at which phase of life ?
- (1) During Intra-uterine life
 - (2) at Puberty
 - (3) During Fertilization
 - (4) During Implantation
162. In human at the end of the second meiotic division the male germ cells convert into the :-
- (1) Primary spermatocytes
 - (2) Spermatids
 - (3) Spermatozoa
 - (4) Secondary spermatocyte
163. Which of the following hormone is release by sertoli cells.
- (1) FSH
 - (2) LH
 - (3) Inhibin
 - (4) Testosterone
164. Seminal plasma of semen contains :-
- (1) Fibrinogen
 - (2) Ca^{+2}
 - (3) Clotting enzyme
 - (4) All of them
165. Epididymis, vas-deference, seminal-vesicle, prostate gland, ejaculatory duct, urethra, penis How-many structures are unpaired ?
- (1) 1
 - (2) 3
 - (3) 4
 - (4) 5
166. Which was the first human being like creature ?
- (1) Dryopithecus
 - (2) Ramapithecus
 - (3) Homo-habilis
 - (4) Homo-sapiens
167. Nature of pituitary hormone which target is Sertoli cell :-
- (1) Steroid Hormone
 - (2) Amino-acid
 - (3) Glycoprotein
 - (4) Cholesterol

168. Interstitial space of Testicular lobule contain :-
- (1) Immuno-competent cells
 - (2) Leydig's-cells
 - (3) Sertoli-cells
 - (4) 1 and 2 both
169. Which of the following helps in motility and maturation of sperm :-
- (1) Epididymis
 - (2) Vas-deference
 - (3) Accessory Reproductive glands
 - (4) All of them
170. In the given figure identify the part labelled as A, B, C & D –



- (1) A → Prostate, B → Vas-deference, C → Seminal vesicle, D → Bulbourethral
 - (2) A → Prostate, B → Seminal vesicle, C → Prostate, D → Bulbourethral
 - (3) A → Seminal vesicle, B → Vas-deference, C → Prostate, D → Bulbourethral
 - (4) A → Vas-deference, B → Seminal vesicle, C → Prostate, D → Bulbourethral
171. Part of fallopian tube which present near to ovary is known as :-
- (1) Infundibulum
 - (2) Isthmus
 - (3) Ampulla
 - (4) Fundus

172. Which is another name of luteal phase of menstrual-cycle?

- (1) Follicular phase
- (2) Estrogenic phase
- (3) Secretory phase
- (4) Pre-ovulatory phase

173. **Statement-I** : The endometrium undergoes cyclic changes during menstrual-cycle.

Statement-II : Change in uterus are induced by change in level of ovarian hormones.

- (1) Both Statement-I and Statement-II are incorrect.
- (2) Both Statement-I and Statement-II are correct.
- (3) Statement-I is correct but Statement-II is incorrect.
- (4) Statement-I is incorrect but Statement-II is correct.

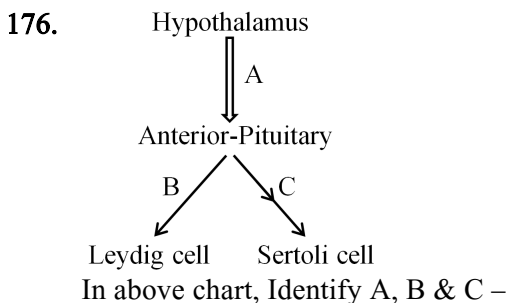
174. Release of sperm from male body is known as :

- (1) Semination
- (2) Spermiation
- (3) Insemination
- (4) Ejaculation

175. How many structures from given below list belong to haploid set of chromosome.

Sperm, spermatid, polar body, spermatogonia, secondary oocyte, oogonia.

- (1) 3
- (2) 4
- (3) 2
- (4) 5



- (1) A → GHRH, B → LH, C → FSH
- (2) A → GnRH, B → FSH, C → LH
- (3) A → GnRH, B → LH, C → FSH
- (4) A → GHRH, B → FSH, C → LH

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177. Which structure carries sperm from seminiferous tubule to vas-efferentia?

- (1) Rete-testis
- (2) Vas-deference
- (3) Ampulla
- (4) Ejaculatory duct

178. Match the column-A with column-B :-

Column-A		Column-B	
A	Fusion of male and female gametes	(i)	Parturition
B	Attachment of blastocyst to the uterine wall	(ii)	Gestation
C	Embryonic development	(iii)	Fertilization
D	Delivery of the baby	(iv)	Implantation

- (1) A–(iv), B–(ii), C–(i), D–(iii)
- (2) A–(iii), B–(iv), C–(i), D–(iii)
- (3) A–(ii), B–(iv), C–(iii), D–(i)
- (4) A–(iii), B–(iv), C–(ii), D–(i)

179. Given below are two statement –

Statement-I : At puberty only 60,000–80,000 primary follicles are left in each ovary.

Statement-II : The secondary follicle is characterised by a fluid filled cavity called antrum.

- (1) Both Statement-I and Statement-II are incorrect.
- (2) Both Statement-I and Statement-II are correct.
- (3) Statement-I is correct but Statement-II is incorrect.
- (4) Statement-I is incorrect but Statement-II is correct.

180. Assertion: Menstruation only occurs if the released ovum is not fertilized.

Reason: During ovulation only one ovum released per menstrual cycle.

- (1) Both assertion & reason are true & the reason is a correct explanation of the assertion.
- (2) Both assertion & Reason are true but reason is not a correct explanation of the assertion.
- (3) Assertion is true but the reason is false.
- (4) Both assertion & reason are false.

181. How many are part of external genitalia in female ?

Cervix, Hymen, Vagina, Clitoris, Labia minora.

- (1) 5 (2) 4 (3) 3 (4) 2

182. Inner side of seminiferous tubule is lined by :

- (1) Spermatogonia and leydig cells.
- (2) Spermatid and leydig cells.
- (3) Spermatogonia and Sertoli cells.
- (4) P.G.C. and white fibrous connective tissue.

183. In primary oocyte of human female, Ist meiotic division is completed :-

- (1) Prior to formation of yellow body
- (2) Prior to ovulation
- (3) Prior to birth
- (4) Prior to puberty

184. Ejaculatory duct is formed by :-

- (1) Urethra and vas-deferens
- (2) Urethra and duct of seminal vesicle
- (3) Vas-deferens and duct of seminal vesicle
- (4) Vas-deferens and epididymis

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185. Select the number of sperms and ova produced by 100 primary spermatocytes and primary oocytes respectively.

- (1) 400 and 400
- (2) 200 and 200
- (3) 400 and 100
- (4) 200 and 50

SECTION-B (BIOLOGY-II)

186. The injection given against the snake venom contains :-

- (1) Antigenic proteins
- (2) Preformed antibodies
- (3) Attenuated pathogen
- (4) All of these

187. A person suffering from leukaemia has :-

- (1) Tumors in Adipose tissue
- (2) Increased number of plasma cells
- (3) Increased number of melanocytes
- (4) Increased number of WBCs

188. Interferons act against viral infection by :-

- (1) Protecting other cells against virus
- (2) Functioning as macrophages
- (3) Providing antibodies
- (4) Directly destroying genetic material of virus

189. A drug substance which interferes with neurotransmitter dopamine is extracted from :-

- (1) Cannabis sativa
- (2) Dhatura
- (3) Erythroxylum coca
- (4) Atropa belladonna

190. After getting into the body of the person HIV virus enters into macrophages where RNA genome of the virus replicates to form viral DNA with the help of ?

- (1) DNA polymerase
- (2) Integrase
- (3) Reverse transcriptase
- (4) Protease

191. Match the column & choose the correct option.

Column-I		Column-II	
(A)	B-lymphocyte	(i)	Primary and secondary immune response carried out by them
(B)	T-lymphocyte	(ii)	Produce army of protein in response to pathogen
		(iii)	Able to differentiate between self and non self
		(iv)	They do not produce antibodies

- (1) A - i, iii ; B - i, iv
- (2) A - i, ii, iii ; B - i, iii, iv
- (3) A - i, ii, iv ; B - ii, iii
- (4) A - i, iv ; B - ii, iii

192. A child was noticed with round head, furrowed tongue palm crease and its physical, psychomotor and mental development is retarded. Identify this disorder :-

- (1) Down syndrome (2) Turner syndrome
- (3) Edward syndrome (4) Cry-du-chat

193. A person has long legs, female like appearance gynecomastia and sterile is suffering from :-

- (1) Down syndrome
- (2) Turner syndrome
- (3) Klinefelter's syndrome
- (4) Super male

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194. Plasmodium enters the human body as (A)..... (infections stage) through the bite of infected female (B)..... mosquito.

	(A)	(B)
(1)	Cryptozoite	<i>Aedes aegypti</i>
(2)	Sporozoite	<i>Anopheles</i>
(3)	Trophozoite	<i>Culex</i>
(4)	Gametocyte	<i>Anopheles</i>

195. Match the columns with regards to Vector and Disease

	Column-I		Column-II
p.	<i>Culex</i>	i	Dengue
q.	<i>Anopheles</i>	ii.	Filariasis
r.	<i>Aedes</i>	iii.	Malaria

- (1) p-i, q-ii, r-iii (2) p-ii, q-iii, r-i
- (3) p-ii, q-i, r-iii (4) p-i, q-iii, r-ii

196. Read the following statements (A-E)



- (A) Gross deformities in genital organs
 - (B) Elephantiasis
 - (C) Vector is male aedes mosquito
 - (D) Slowly developing chronic inflammation usually of blood vessels of lower limbs
 - (E) It is caused by *Wuchereria bancrofti*.
- Statements incorrect about given diagram are :-

- (1) A and C
- (2) B and C
- (3) C and D
- (4) D and E

197. Match the following column 'A' and 'B' ?

(A)		(B)	
(i)	Tears	(a)	Physiological barrier
(ii)	NK cell	(b)	Cellular barrier
(iii)	Skin	(c)	Physical barrier
(iv)	MALT	(d)	Gastro Intestinal tract

	(i)	(ii)	(iii)	(iv)
(1)	a	b	c	d
(2)	d	a	b	c
(3)	b	a	c	d
(4)	a	d	c	b

198. How many of the following statements are correct about given diagram ?

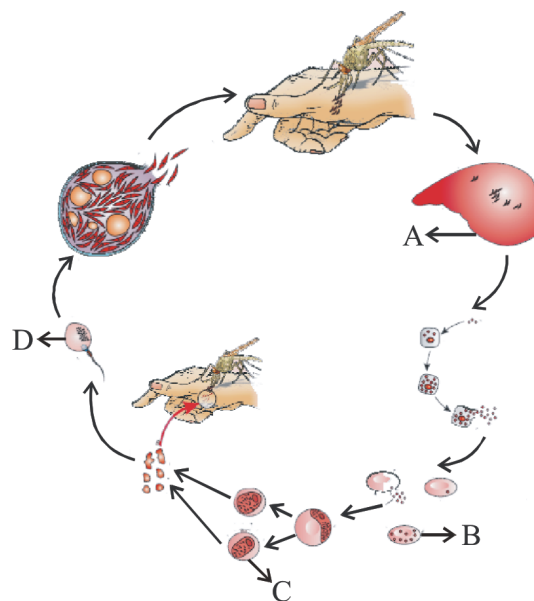


- A. Microsporum, Trichophyton and Epidermophyton are responsible for this
- B. Allergic disease
- C. Heat & moisture help these to grow
- D. It is caused by Wuchereria bancrofti

- (1) Four
- (2) One
- (3) Two
- (4) Three

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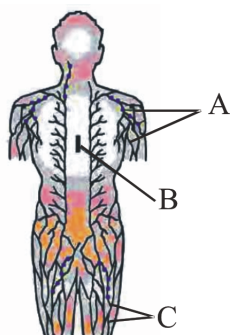
199. Identify the stages marked as A, B, C and D in the life cycle of *Plasmodium*:-



Options :-

	A	B	C	D
(1)	Sporozoites in spleen	Sexual reproduction of Plasmodium in RBC	Gametocytes in RBC	Fertilization of gamete in mosquito's stomach
(2)	Sporozoites in liver	Asexual reproduction of Plasmodium in RBC	Gametocytes in RBC	Fertilization of gamete in mosquito's gut
(3)	Sporozoites in liver	Sexual reproduction of Plasmodium in RBC	Gametocytes in RBC	Fertilization of gamete in mosquito's salivary glands
(4)	Sporozoites in kidney	Sexual reproduction of Plasmodium in RBC	Gametocytes in RBC	Fertilization of gamete in mosquito's gut

200. Identify the structures A, B and C related to the immune system:-



Options	A	B	C
(1)	T-lymphocytes mature here	Thymus	Lymphatic vessels
(2)	Lymph nodes	T-lymphocytes mature here	Lymphatic vessels
(3)	Thymus	Lymph nodes	Lymphatic vessels
(4)	Lymphatic vessels	Lymph nodes	Thymus



**QUESTIONS BREAK-UP CHART FOR MINOR TEST
FROM CURRENT & PREVIOUS UNITS
(Session : 2023 - 2024)**

ENTHUSE ADV. COURSE : MEC

MINOR TEST NO. & DATE	CURRENT UNIT	PREVIOUS UNIT (COMPLETE TOPIC)	
04 18-06-2023	SYLLABUS COVERED FROM 22-05-2023 TO 17-06-2023	P	ELECTROSTATICS AND CURRENT ELECTRICITY
		C	SOLUTION
		B	ORIGIN, EVOLUTION

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